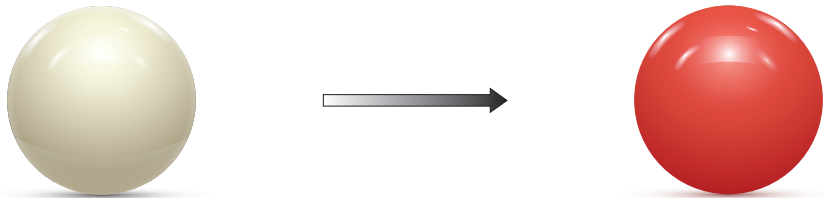
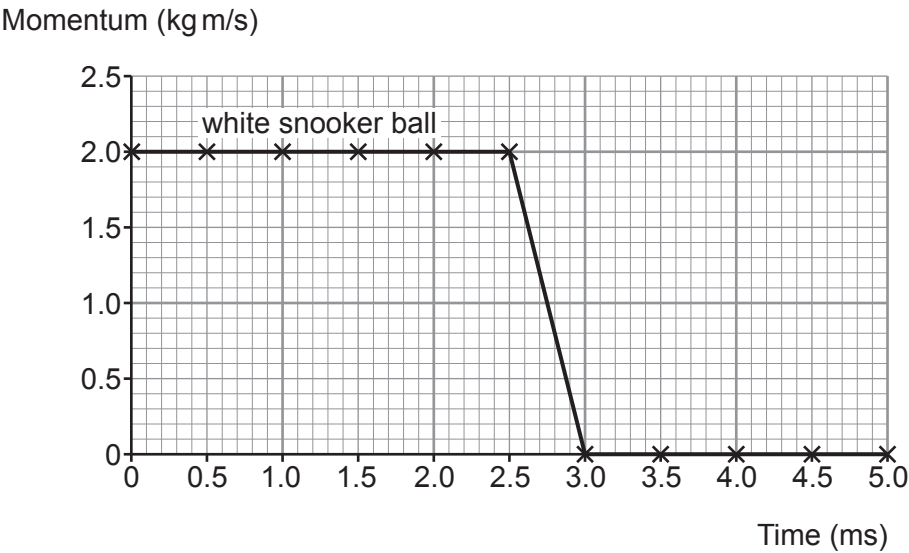


Answer **all** questions.

1. A white snooker ball, rolling to the right, collided with a **stationary** red snooker ball at a time of 2.5 ms.  
Each snooker ball has a mass 0.16 kg.



The graph below shows the momentum of the **white snooker ball** before and after the collision.



- (a) Use information from the graph to answer the following questions.

- (i) State the initial momentum of the white snooker ball. [1]

initial momentum = ..... kg m/s



- (ii) Use the equation:

$$\text{initial velocity} = \frac{\text{initial momentum}}{\text{mass}}$$

to calculate the initial velocity of the white snooker ball.

[2]

initial velocity = ..... m/s

- (iii) The collision takes a time of 0.5 ms. State this time in seconds.

[1]

time = ..... s

- (iv) Use the equation:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

to calculate the resultant force on the white snooker ball during the collision.

[2]

resultant force = ..... N

- (b) (i) Underline the correct word or phrase from each of the brackets below which correctly completes the sentence.

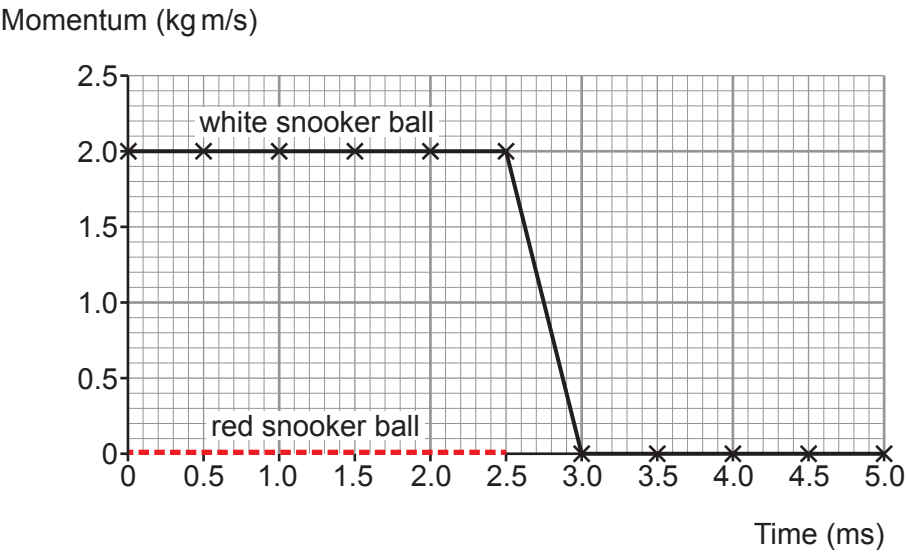
[2]

The law of conservation of momentum states that the total momentum before a collision is (**less than** / **equal to** / **greater than**) the total momentum after a collision provided (**no** / **small** / **large**) external forces act.



Examiner  
only

- (ii) The momentum of the red snooker ball, between 0.0 and 2.5 ms, has been added to the original graph. It is shown as a red dotted line.  
**Complete the graph** below to show the momentum of the red snooker ball from 2.5 ms to 5.0 ms. [2]



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05

